

5.4 Common mistakes in the use of colour in data visualisations ↕

□ Watch: Purposeful use of colour – part 3 [14.28 mins]

Watch the following video, which explains how colour can be used inappropriately and how this detracts from the message in a visualisation.

This is useful as it provides understanding of techniques to avoid when using colour in a visualisation.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138645?wrap=1>

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Download the following slides for note taking: [ICT702 Purposeful use of colour – part 3 \(https://learn.usc.edu.au/courses/35242/files/3138663?wrap=1\)](https://learn.usc.edu.au/courses/35242/files/3138663?wrap=1)

Common mistakes made with colour

There are a number of common mistakes people make when using colour in data visualisations. Descriptions of some of the major ones are listed here.

Unnecessary colour

Colour should only be used when it communicates some information that other chart elements do not. The use of multiple colours increases cognitive load and therefore should be avoided.

Excessive colour

The amount of information that can be communicated through colour is limited; therefore, care should be taken when using colour.

Insufficient colour contrast

If colour is used to distinguish between chart elements, it is important to make sure that it is easy to differentiate between the colours. If the colours can be confused, the cognitive load required by the audience to understand the message of the visualisation is increased.

Inconsistency across related charts

Inconsistency in colour across related charts causes confusion and increased cognitive load for the audience.

Neglecting colour-blindness

Colour-blindness is the reduced ability to accurately perceive certain colours. It is caused by at least one of the types of cones in the retina being insensitive to the wavelength that that type of cone should sense. Red-green colour-blindness, which is most common, results in an inability to perceive (to some degree) the difference between red and green. It is more prevalent in men than women. Blue-yellow colour-blindness may also occur. A small percentage of the population is completely colour-blind, in which case they only see shades of grey. If you ignore the possibility of members of the audience being colour-blind, some members of the audience may not understand the message you are trying to provide.

Not considering the mode of delivery

When choosing colours, the mode of delivery of a presentation is important. Presentations that will be projected on to a screen should make good use of use colour contrast, high saturation and luminance, sharp outlines and thick lines. By contrast, printed mediums should use colours with less saturation, less contrast, lower luminance and softer outlines. It is important to remember that colours will vary from monitor to monitor, projector to projector, printer to printer and any combination of those items. Always check that an output is as you intended.